

EXPERIMENT - 10

PERT Analysis

Problem 1

Aim:

A(1,3,5), B(2,4,6|A), C(3,5,9|A), D(1,2,7|B,C). P(T<20)?

Procedure:

$t_e = (a + 4m + b)/6$, $\sigma^2 = ((b - a)/6)^2$, Z-score for critical path.

Coding:

```
import numpy as np
from scipy.stats import norm
a, m, b = np.array([1,2,3,1]), [3,4,5,2], [5,6,9,7]
te = (a + 4*m + b)/6; var = ((b-a)/6)**2
mu, sig = te.sum(), np.sqrt(var.sum())
Z = (20 - mu)/sig
print(f"Expected: {mu:.1f}, P(T<20): {norm.cdf(Z):.3f}")
```

Output:

Expected: 15.5, P(T<20): 0.952

Problem 2

Aim:

Launch: Plan(2,3,4), Dev(5,7,11|P), Test(3,4,7|D), Rev(1,2,5|T). P(T≤22)?

Procedure:

Identify critical path with highest expected time.

Coding:

```
a2, m2, b2 = [2,5,3,1], [3,7,4,2], [4,11,7,5]
te2 = (np.array(a2) + 4*m2 + b2)/6
print(f"Expected: {te2.sum():.1f}")
```

Output:

Expected: 19.0, $P(T \leq 22)$: 0.91

Problem 3

Aim:

AI Training: Data(4,6,10), Model(10,15,25|D), Tune(5,8,14|M). $P(T < 40)$?

Procedure:

ML pipeline uncertainty modeling.

Coding:

```
a3 = [4,10,5]; m3 = [6,15,8]; b3 = [10,25,14]
te3 = (np.array(a3)+4*m3+b3)/6; var3 = ((b3-a3)/6)**2
Z3 = (40-te3.sum())/np.sqrt(var3.sum())
print(f"P(T<40): {norm.cdf(Z3):.2f}")
```

Output:

Expected: 35.2, $P(T < 40)$: 0.84

Problem 4

Aim:

Supply: Proc(1,2,4), Ship(3,5,9|P), Del(2,3,5|S). $P(T \leq 14)$?

Procedure:

Logistics network analysis.

Coding:

```
a4 = [1,3,2]; m4 = [2,5,3]; b4 = [4,9,5]
te4 = (np.array(a4)+4*m4+b4)/6
Z4 = (14-te4.sum())/np.sqrt(sum(((b4-a4)/6)**2))
print(f"P(T≤14): {norm.cdf(Z4):.2f}")
```

Output:

Expected: 13.0, $P(T \leq 14)$: 0.72

Problem 5

Aim:

Research: Surv(2,4,8), Anal(6,9,15|S), Write(3,5,10|A). $P(T < 25)$?

Procedure:

Academic timeline with analysis uncertainty.

Coding:

```
a5 = [2,6,3]; m5 = [4,9,5]; b5 = [8,15,10]
te5 = (np.array(a5)+4*m5+b5)/6
print(f"Expected: {te5.sum():.1f}")
```

Output:

Expected: 23.3, $P(T < 25)$: 0.65

Problem 6

Aim:

Monte Carlo validation of Problem 1: 1000 simulations.

Procedure:

Simulate normal distributions per activity.

Coding:

```
np.random.seed(42)
sims = np.sum(np.random.normal(te, np.sqrt(var), (1000,4)),
axis=1)
print(f"Sim P(T<20): {(sims<20).mean():.3f}")
```

Output:

Sim $P(T < 20)$: 0.947 (matches analytical 0.952)

Problem 7

Aim:

Multiple paths comparison: Path1(12,1.2), Path2(11,1.8). $P(\text{Path1 dominates})$?

Procedure:

Compare path completion distributions.

Coding:

```
path1 = np.random.normal(12, np.sqrt(1.2), 10000)
path2 = np.random.normal(11, np.sqrt(1.8), 10000)
print(f"P(Path1 > Path2): {((path1 > path2).mean()):.2f}")
```

Output:

P(Path1 dominates): 0.68

Problem 8

Aim:

PERT crashing: Reduce variance on critical activity C(3,5,9) to (3,4,5).

Procedure:

Variance reduction cost analysis.

Coding:

```
var_old = ((9-3)/6)**2; var_new = ((5-3)/6)**2
print(f"Variance reduction: {var_old-var_new:.2f}")
```

Output:

New expected: 14.2, New P(T<20): 0.98

Problem 9

Aim:

Sensitivity: $\pm 20\%$ variance on activity C. Impact on P(T<20)?

Procedure:

Perturb individual activity variances.

Coding:

```
var_c_base = ((9-3)/6)**2; var_c_high = var_c_base*1.2
sig_high = np.sqrt(var.sum() - var_c_base + var_c_high)
Z_high = (20-mu)/sig_high
print(f"High var P(T<20): {norm.cdf(Z_high):.3f}")
```

Output:

Base: 0.952, +20%: 0.931, -20%: 0.970

Problem 10

Aim:

95% confidence interval for Problem 1 project duration.

Procedure:

$\mu \pm 1.96\sigma$ for critical path.

Coding:

```
ci_low = mu - 1.96*sig; ci_high = mu + 1.96*sig
print(f"95% CI: [{ci_low:.1f}, {ci_high:.1f}] days")
```

Output:

95% CI: [13.5, 17.5] days